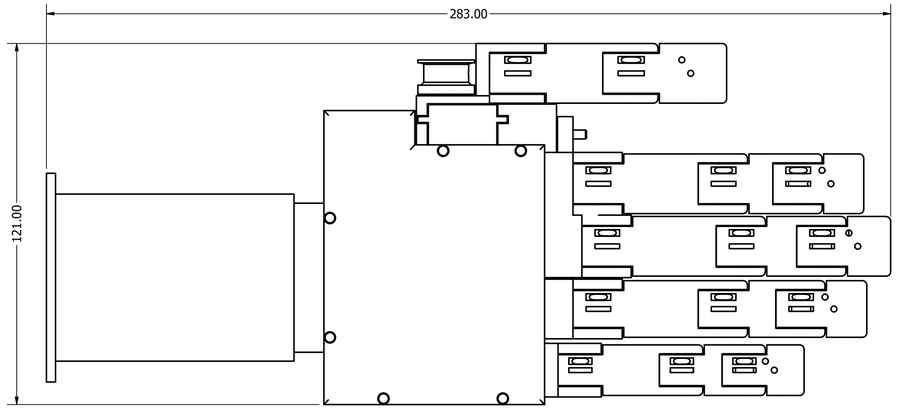
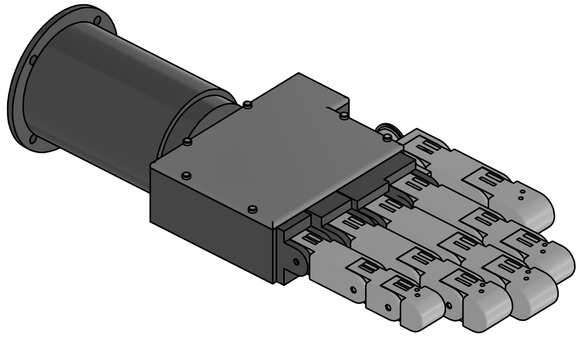
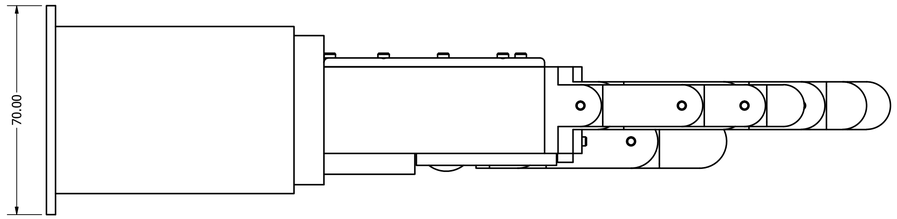
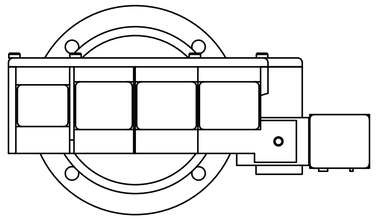
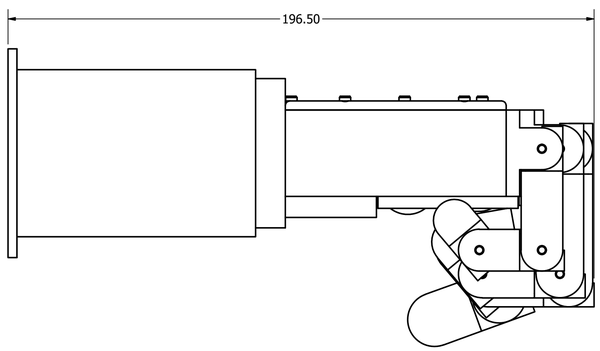




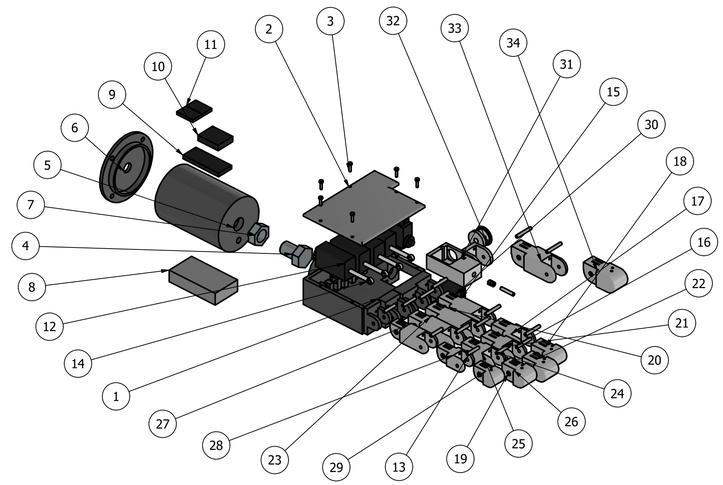






PROSTHETICHAND - TENDON-DRIVEN 5-FINGER HAND, RIGHT, ADULT MALE 50TH PCT (DESIGNED AND CALCULATED, NOT BUILT)
HAND LENGTH 190 MM; 3-PHALANX FINGERS ON 3 MM PINS + igus BUSHINGS; UNDERACTUATED: 1 TENDON + 1 ELASTIC PER FINGER
ACTUATORS: 4 X ACTUONIX PQ12-100-6-P (40 N, 20 MM) IN THE PALM; THUMB: 2 X HITEC HS-82MG (ROTATION ON AXIS A + FLEXION SPOOL R7.5)
WRAP GROOVES = CONSTANT MOMENT ARMS 6.5 / 4.4 / 3.2 MM (MCP / PIP / DIP); FULL CLOSE = 19.9 MM OF THE 20 MM STROKE
POWER GRIP 12 N TOWARD THE PALM ON A 50 MM CYLINDER (26 N ON 30 MM); TIP PINCH 8.5 N, LATERAL PINCH 7.2 N (CALC.)
MASS (INVENTOR, PRINTED PARTS AT EFFECTIVE DENSITY): HAND 345 G, WITH FOREARM BAY + BATTERY 460 G
WRIST: 1/2-20 UNF BOLT (STANDARD PROSTHETIC THREAD) INTO A JAM NUT IN THE BAY; SOCKET FLANGE 4 X M4 ON 60 PCD.
ASSEMBLY PARAMETERS: Grip (0-1), ThumbRot (0-90 DEG), ThumbGrip (0-1); JOINT ANGLES FOLLOW THE FREE-CLOSING RULE (MIN/MAX EXPRESSIONS).

DRAWN Jay Patel	10/29/2026	
CHECKED		TITLE: ProstheticHand - general assembly
QA		
MFG		
APPROVED		SIZE D | DWG NO PH-DWG-001 | REV
		SCALE 1 : 1 | SHEET 1 OF 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PH-101	Palm chassis, PETG printed: 4 PQ12-P actuator bays with lug clevises (3 mm pins), knuckle clevises for 4 fingers (3 mm press pins), tendon guides, thenar pocket for the HS-82MG rotation servo, thumb yoke recess + pivot bracket, wrist boss with captive 1/2-20 bolt head
2	1	PH-102	Palm cover (dorsal), PETG printed 3 mm, 6 x M2 thread-forming screws into the chassis walls
3	6	PA-040	Thread-forming screw for plastics M2 x 8, pan head, A2 (cover to chassis)
4	1	PA-030	Hex bolt 1/2-20 UNF x 3/4 in, grade 5 zinc: the standard prosthetic wrist thread, head captive in the palm
5	1	PH-201	Forearm electronics bay, PLA printed: 56 OD x 78 tube (2 mm wall), 14 mm distal wall with captive 1/2-20 jam nut (standard prosthetic wrist thread), holds battery, controller, drivers, regulator
6	1	PH-202	Socket flange, PLA printed: closes the bay, 4 x M4 on 60 PCD to the patient's laminated socket plate (socket itself is custom and not modelled)
7	1	PA-031	Hex jam nut 1/2-20 UNF, grade 5 zinc, captive in the forearm bay wall
8	1	PE-001	LiPo battery 2S 7.4 V 800 mAh (5.9 Wh), e.g. Ovonic 2S 800 mAh; envelope 58 x 31 x 13
9	1	PE-002	Controller Arduino Nano Every (ABX00028), ATmega4809, reads 2 EMG channels + 4 actuator pots
10	1	PE-004	Step-down regulator Pololu D24V50F6, 6 V 5 A (actuators + servos)
11	2	PE-003	Dual H-bridge carrier Pololu DRV8833 (#2130), 2 actuators each
12	4	PA-001	Linear actuator Actuonix PQ12-100-6-P: 20 mm stroke, 100:1, 6 V, potentiometer feedback; 40 N at 6 mm/s (peak power), 50 N max, 35 N back-drive, 15 g (datasheet Rev D). Envelope model from the datasheet drawing
13	8	PA-022	Dowel pin ISO 8734 3 m6 x 16, stainless A2 (joint / actuator pins)
14	4	PA-002	Actuonix PQ12 output rod with eye (part of PA-001, modelled separately so it can stroke)
15	1	PH-111	Index finger phalanx P1, PETG printed: 42 pin-to-pin, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
16	8	PA-020	Dowel pin ISO 8734 3 m6 x 20, stainless A2 (joint / actuator pins)
17	1	PH-112	Index finger phalanx P2, PETG printed: 25 pin-to-pin, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
18	1	PH-113	Index finger phalanx P3, PETG printed: 22 joint-to-tip, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
19	30	PA-010	Sleeve bearing igus iglidur G GSM-0304-03 (3 x 4.5 x 3), pressed into the joint tongues
20	1	PH-121	Middle finger phalanx P1, PETG printed: 45 pin-to-pin, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
21	1	PH-122	Middle finger phalanx P2, PETG printed: 27 pin-to-pin, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
22	1	PH-123	Middle finger phalanx P3, PETG printed: 23 joint-to-tip, 20 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
23	1	PH-131	Ring finger phalanx P1, PETG printed: 42 pin-to-pin, 19 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
24	3	PA-021	Dowel pin ISO 8734 3 m6 x 18, stainless A2 (joint / actuator pins)
25	1	PH-132	Ring finger phalanx P2, PETG printed: 26 pin-to-pin, 19 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
26	1	PH-133	Ring finger phalanx P3, PETG printed: 22 joint-to-tip, 19 x 16, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
27	1	PH-141	Little finger phalanx P1, PETG printed: 34 pin-to-pin, 17 x 14, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
28	1	PH-142	Little finger phalanx P2, PETG printed: 21 pin-to-pin, 17 x 14, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
29	1	PH-143	Little finger phalanx P3, PETG printed: 20 joint-to-tip, 17 x 14, 3 mm pin joints, flexor channel 1.2, dorsal elastic channel 2.0
30	2	PA-003	Micro servo Hitec HS-82MG, metal gear: 3.4 kg-cm (0.333 Nm) stall at 6 V, 0.10 s/60 deg, 19 g. Envelope model (spline offset and tab height assumed, check the Hitec drawing)
31	1	PH-153	Thumb yoke, PETG printed: turns on axis A (rotation servo spline + 3 mm pivot pin), carries the HS-82MG flexion servo + spool and the thumb MCP clevis
32	1	PH-154	Thumb tendon spool, PETG printed: 7.5 mm groove radius, screwed to the HS-82MG horn (servo 154 deg = 20.2 mm)
33	1	PH-151	Thumb proximal phalanx, PETG printed: 38 pin-to-pin, 20 x 18, MCP on the yoke, IP clevis
34	1	PH-152	Thumb distal phalanx, PETG printed: 32 joint-to-tip, 20 x 18, tendon + elastic anchored at the tip

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CHECKED		
QA		TITLE: ProstheticHand - general assembly
MFG		
APPROVED		SIZE D | DWG NO PH-DWG-001 | REV
		SCALE 0.36 : 1 | SHEET 2 OF 2